

# BTIH Mobile App — Project Status Report

Bahria Town International Hospital Karachi · Patient Mobile Application (Flutter)

Repository: `bti_h_andriod_app` | Platform: Flutter (Dart), Android-first | Backend: .NET Core Web API (Swagger) + Oracle HMIS

Report generated: 11 Aug 2026 | Basis: full source review of `lib/` (37 Dart files), build config, and manifest.

## 1. How to read this report

Every feature/module is classified into one of four states:

- **COMPLETED** — Built and wired end-to-end (UI + service + backend call), usable.
- **PARTIAL** — Works but with gaps: missing sub-cases, commented-out pieces, or rough edges.
- **INCOMPLETE** — Started (some code exists) but not usable / not reachable in the app.
- **UNTOUCHED** — Not started, placeholder, or an abandoned dependency with no working code.

Note: "usable" assumes the app is pointed at a running API (see the connectivity setup in `docs/ONBOARDING.md`). Backend endpoint correctness is out of scope for this client-side review.

## 2. Executive summary



The app's **core patient journey is functionally complete**: launch → browse doctors → book appointment → log in → view profile, appointments, discharge history, and download/share PDF reports. Authentication (login, forgot-password via OTP, logout) is implemented end-to-end.

The main weaknesses are **(a) security & session robustness** (no token/JWT auth; login state is largely in-memory), **(b) a few half-finished features** (bill categories has backend code but no screen; guest-appointment display is commented out; some report categories are disabled), and **(c) abandoned scaffolding** (Firebase/Google-Sign-In dependencies with no code, several fully-commented legacy files, and the default test still present and broken).

**Overall maturity:** A working *demo/MVP-grade* app. It can be shown and used on a LAN against the dev API. It is *not yet production-hardened* (security, session persistence, error handling consistency, tests, and cleanup remain).

### 3. Feature status — master table

| Feature / Module                                      | Status    | Notes & evidence                                                                                                                                                                                    |
|-------------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| App launch & splash                                   | COMPLETED | Animated splash ( <code>splash_screen.dart</code> ) → navigates to Dashboard as guest after ~3s.                                                                                                    |
| Dashboard hub + guest gating                          | COMPLETED | Bottom nav, feature tiles, and a "Login Required" dialog that redirects guests to login ( <code>dashboard_screen.dart</code> ).                                                                     |
| Login (phone + password)                              | COMPLETED | POST <code>/api/Auth/login</code> ; parses <code>mR_NO</code> → navigates to dashboard as logged-in. Auth class lives inside <code>login_screen.dart</code> .                                       |
| Forgot password (OTP flow)                            | COMPLETED | Multi-step UI: verify phone → <code>send-otp</code> → <code>verify-otp</code> → <code>updatePassword</code> . All wired with resend + validation.                                                   |
| Logout                                                | COMPLETED | Confirmation dialog, clears <code>SharedPreferences</code> , resets state ( <code>dashboard_screen.dart</code> <code>_logout</code> ).                                                              |
| Doctors list                                          | COMPLETED | Paginated infinite scroll + specialization filter dropdown ( <code>doctors_list_screen.dart</code> + <code>DoctorService</code> + <code>SpecializationService</code> ).                             |
| Doctor schedule + booking                             | COMPLETED | Guest booking (saved to local SQLite + server challan) and logged-in booking (with phone verify), self/relative options ( <code>doctor_schedule_screen.dart</code> + <code>BookingService</code> ). |
| Appointments (current/past)                           | COMPLETED | Fetches <code>/api/Patient/appointments/{mrNo}</code> , tabs + status colors ( <code>AppointmentsInfoScreen.dart</code> ).                                                                          |
| Patient profile                                       | COMPLETED | Fetches <code>/api/Patient?MR_NO=</code> , builds profile from latest visit ( <code>patient_profile_screen.dart</code> ).                                                                           |
| Discharge history                                     | COMPLETED | Paginated via Dio ( <code>discharge_history_screen.dart</code> + <code>DischargeHistoryService</code> ).                                                                                            |
| Report history + PDF view/share                       | COMPLETED | Generates PDF ( <code>GenerateReports</code> ), views with Syncfusion viewer, downloads to file, shares via <code>share_plus</code> ( <code>patient_report_history_screen.dart</code> ).            |
| Discharge report (PDF)                                | COMPLETED | <code>DischargeReportService</code> (Dio, PDF bytes, content-type validation).                                                                                                                      |
| Reports screen<br>(lab/gastro/radiology/prescription) | PARTIAL   | PDF generation works, but several direct per-category fetch endpoints are commented out ( <code>reports_screen.dart</code> lines ~85–162). Relies on the generic generate endpoint.                 |

| Feature / Module                       | Status     | Notes & evidence                                                                                                                                                                                                                                                                    |
|----------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Guest appointment display              | PARTIAL    | Guest bookings are <i>saved</i> to SQLite, but reading them back for display is commented out in <code>AppointmentsInfoScreen.dart</code> (~line 672). So a guest can't see their saved bookings.                                                                                   |
| Session persistence ("stay logged in") | PARTIAL    | Inconsistent: <i>logout</i> clears <code>SharedPreferences</code> , but <i>login</i> never saves the session there — MR/name are passed via constructor and kept in memory. App restart drops you back to guest.                                                                    |
| Local SQLite (guest appointments)      | PARTIAL    | Insert works and is used; read/display path is disabled (see above). Delete/clear helpers exist but unused.                                                                                                                                                                         |
| Code quality / analyzer                | PARTIAL    | ~100 analyzer issues remain (mostly <code>avoid_print</code> and <code>use_build_context_synchronously</code> ). No errors — builds fine. 104 were already auto-fixed earlier.                                                                                                      |
| Bill categories                        | INCOMPLETE | <code>BillCategoryService</code> + <code>BillCategory</code> model exist and compile, but there is <b>no screen</b> ( <code>bill_category_screen.dart</code> is missing; only commented references in <code>main.dart</code> ). Not reachable in the app. Also uses a hardcoded IP. |
| Automated tests                        | INCOMPLETE | Only the default template <code>test/widget_test.dart</code> exists, and it tests a non-existent counter — it will <b>fail</b> if run. Effectively no test coverage.                                                                                                                |
| Messages / chat                        | UNTOUCHED  | Only a "Messages feature coming soon" comment in <code>dashboard_screen.dart</code> . No implementation.                                                                                                                                                                            |
| Firebase integration                   | UNTOUCHED  | <code>firebase_core</code> / <code>firebase_auth</code> in <code>pubspec.yaml</code> , but no <code>Firebase.initializeApp()</code> and no usage anywhere. Dead dependency.                                                                                                         |
| Google / social sign-in                | UNTOUCHED  | <code>google_sign_in</code> present; only commented stubs in <code>auth_service.dart</code> . Not implemented.                                                                                                                                                                      |
| Token / JWT authorization              | UNTOUCHED  | No auth token. After login, endpoints are called with the MR number in the URL and no <code>Authorization</code> header — a security gap (see §6).                                                                                                                                  |
| Home screen                            | UNTOUCHED  | <code>home_screen.dart</code> is a "Welcome to the Home Screen" placeholder; not used in navigation.                                                                                                                                                                                |
| WebView report viewer                  | UNTOUCHED  | <code>services/webview_flutter.dart</code> is entirely commented; the <code>webview_flutter</code> package isn't even in <code>pubspec.yaml</code> . Superseded by the Syncfusion PDF viewer.                                                                                       |
| iOS / Web / Desktop targets            | UNTOUCHED  | Platform folders are auto-generated but unconfigured/untested. The app is Android-only in practice.                                                                                                                                                                                 |

## 4. Module-by-module detail

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### 4.1 Authentication & session

**Completed:** phone+password login, OTP-based forgot-password (verify phone → send OTP → verify → set new password), and logout with confirmation.

**Gaps:** the real auth code sits inside `login_screen.dart` rather than a dedicated service; the separate `auth_service.dart` is legacy and fully commented. Login state is not persisted (in-memory), so users are logged out on every app restart even though logout clears preferences. There is no token/JWT — subsequent API calls identify the patient by MR number in the URL.

### 4.2 Appointments

**Completed:** browsing doctors (paginated + filter), viewing a doctor's schedule, booking for self/relative/guest, and viewing current vs past appointments. Booking for logged-in users writes to the server; guest bookings write to the server *and* local SQLite.

**Gaps:** the guest's locally-saved appointments are never displayed back (the read path in `AppointmentsInfoScreen` is commented out). Some success dialogs/snackbars are commented. A hardcoded phone-verify URL ( `172.16.40.10:8080` ) bypasses the central config.

### 4.3 Reports (PDF)

**Completed:** report history list, on-device PDF generation/download, in-app viewing (Syncfusion), and sharing ( `share_plus` ); discharge report generation.

**Gaps:** in `reports_screen.dart`, direct per-category fetches (lab/gastro/radiology/prescription) are commented out; the screen leans on the generic generate endpoint. Some "share coming soon" remnants remain.

### 4.4 Profile & discharge

**Completed:** patient profile from latest visit; paginated discharge history; discharge report PDF. Solid loading/error states in these screens.

### 4.5 Billing

**Incomplete:** the service and model are written (and even handle a self-signed cert), but there is no UI screen and no navigation entry, so the feature can't be used. It also points at a separate hardcoded server.

## 5. Dead / legacy files (safe to clean up)

| File                                              | Status     | Recommendation                                                                                                      |
|---------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------|
| <code>lib/services/auth_service.dart</code>       | UNTOUCHED  | Legacy/commented. Either delete, or refactor the inline auth logic out of <code>login_screen.dart</code> into here. |
| <code>lib/services/patients_service.dart</code>   | UNTOUCHED  | Fully commented. Delete.                                                                                            |
| <code>lib/services/webview_flutter.dart</code>    | UNTOUCHED  | Fully commented; package not installed. Delete.                                                                     |
| <code>lib/services/url_launcher.dart</code>       | UNTOUCHED  | Fully commented helper. Delete ( <code>url_launcher</code> is used inline elsewhere).                               |
| <code>lib/models/user_model.dart</code>           | UNTOUCHED  | <code>User</code> class defined but not used anywhere. Adopt it in the login flow, or delete.                       |
| <code>lib/screens/home_screen.dart</code>         | UNTOUCHED  | Placeholder screen. Delete or repurpose.                                                                            |
| <code>lib/services/apointment_service.dart</code> | PARTIAL    | Misnamed (it's actually the <code>Appointment</code> model, not a service). Rename/move to <code>models/</code> .   |
| <code>test/widget_test.dart</code>                | INCOMPLETE | Default counter test — broken. Replace with real tests or remove.                                                   |

## 6. Technical debt & risks (prioritized)

**Security (high):** No token/JWT auth; patient data is fetched using the MR number in the URL with no authorization header, so any client that knows/guesses an MR number could pull records directly from the API. The backend also ships plaintext DB credentials in `appsettings.json`. These should be addressed before any production/public release.

- HIGH Auth tokens:** introduce JWT/bearer auth end-to-end (API + app).
- HIGH Config & secrets:** stop committing credentials; use environment/secret storage; rotate the exposed DB password.
- MED Hardcoded IPs** bypassing `ApiConfig.baseUrl` in `bill_category_service.dart` ( `172.16.40.56` ) and `doctor_schedule_screen.dart` ( `172.16.40.10` ). Unify to one config.
- MED Session persistence:** save login to `SharedPreferences` and restore on splash; make logout/login symmetric.
- MED Context-after-await** warnings (potential crashes) across several screens — add `if (!mounted) return;` guards.
- LOW Cleanup:** remove dead files & unused dependencies (Firebase/Google if not planned); fix `print()` usage; fix naming lints.
- LOW Tests:** add at least smoke tests for services/models; fix the broken default test.

## 7. Recommended roadmap

| Phase                 | Goal                                      | Work items                                                                                                                                                                   |
|-----------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1 — Stabilize</b> | Make current features robust & consistent | Unify all URLs to <code>ApiConfig</code> ; persist session on splash; add <code>mounted</code> guards; finish guest-appointment display; re-enable/verify report categories. |
| <b>P2 — Secure</b>    | Production-grade auth & secrets           | JWT/bearer auth (API + app); move backend secrets out of source; enforce HTTPS with a trusted cert.                                                                          |
| <b>P3 — Complete</b>  | Finish half-built features                | Build the Bill Categories screen (or drop the code); decide on Messages/social login (implement or remove deps).                                                             |
| <b>P4 — Quality</b>   | Maintainability                           | Remove dead files; extract auth into a service; add tests; drive analyzer issues to zero.                                                                                    |

## 8. Appendix — source inventory ( [lib/](#) )

| Path                                                                                 | Type    | Role                                                               |
|--------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------|
| <code>main.dart</code>                                                               | entry   | App root → Splash.                                                 |
| <code>utils/ip_file.dart</code>                                                      | config  | <b>Central API base URL</b><br>( <code>ApiConfig.baseUrl</code> ). |
| <code>utils/database_helper.dart</code>                                              | util    | SQLite (guest appointments).                                       |
| <code>utils/date_formatter.dart</code> ,<br><code>utils/validation_mixin.dart</code> | util    | Date formatting; form validators.                                  |
| <code>screens/splash_screen.dart</code>                                              | screen  | Launch animation → dashboard.                                      |
| <code>screens/dashboard_screen.dart</code>                                           | screen  | Hub, nav, login gating, logout.                                    |
| <code>screens/login_screen.dart</code>                                               | screen  | Login + OTP + password reset (contains inline auth service).       |
| <code>screens/doctors_list_screen.dart</code>                                        | screen  | Paginated doctors + filter.                                        |
| <code>screens/doctor_schedule_screen.dart</code>                                     | screen  | Schedule + booking (guest/self/relative).                          |
| <code>screens/AppointmentsInfoScreen.dart</code>                                     | screen  | Current/past appointments (misnamed file).                         |
| <code>screens/patient_profile_screen.dart</code>                                     | screen  | Patient profile.                                                   |
| <code>screens/discharge_history_screen.dart</code>                                   | screen  | Discharge list.                                                    |
| <code>screens/reports_screen.dart</code>                                             | screen  | Report categories + PDF generate.                                  |
| <code>screens/patient_report_history_screen.dart</code>                              | screen  | Report history + view/share PDF.                                   |
| <code>screens/home_screen.dart</code>                                                | screen  | Unused placeholder.                                                |
| <code>services/doctors_service.dart</code>                                           | service | Doctors + schedule.                                                |
| <code>services/specialization_service.dart</code>                                    | service | Specializations (used by filter).                                  |
| <code>services/booking_service.dart</code>                                           | service | Insert appointment (challan).                                      |
| <code>services/discharge_history_service.dart</code>                                 | service | Discharge history (Dio).                                           |
| <code>services/discharge_report_service.dart</code>                                  | service | Discharge report PDF (Dio).                                        |
| <code>services/patient_report_service.dart</code>                                    | service | Report history.                                                    |
| <code>services/bill_category_service.dart</code>                                     | service | Bill data (no UI; hardcoded IP).                                   |

| Path                                                                                                | Type    | Role                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| services/auth_service.dart ,<br>patients_service.dart ,<br>webview_flutter.dart , url_launcher.dart | service | Legacy/commented — unused.                                                                                                                                                                   |
| models/*.dart                                                                                       | models  | patient, doctors, doctor_schedule,<br>discharge_history, patient_report,<br>specialization, bill_category,<br>local_appointment, user (unused),<br>appointment (in apointment_service.dart). |

Generated from a manual code review. Counts in the executive summary group related items (e.g. "Local SQLite" and "Guest appointment display" both reflect the same partial area). For live status, run `flutter analyze` and search the codebase for `// TODO` , commented `Navigator` blocks, and "coming soon" markers.